Definition 1
Two triangles are said to be similar if their corresponding sides are proportional.

Definition 2
The triangles are equiangular if the corresponding angles are equal.

Illustration Two triangles, ΔXYZ and ΔLMN are similar because the corresponding angles are equal.

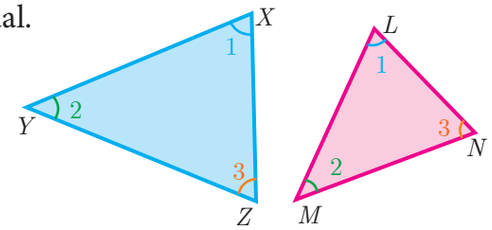


Fig. 4.14

Note

(i) A pair of equiangular triangles are similar.

(ii) If two triangles are similar, then they are equiangular.

(i) $\angle X = \angle L, \angle Y = \angle M, \angle Z = \angle N$ (by angles) (ii) $\frac{XY}{LM} = \frac{YZ}{MN} = \frac{XZ}{LN}$ (by sides)

Here the vertices $X,\ Y,\ Z$ correspond to the vertices $L,\ M,\ N$ respectively. Thus in symbol $\Delta XYZ \sim \Delta LMN$

Example 4.1 Show that $\Delta PST \sim \Delta PQR$

(i)

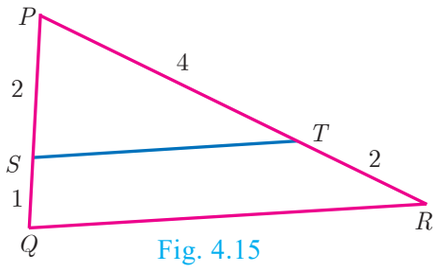


Fig. 4.15

(ii)

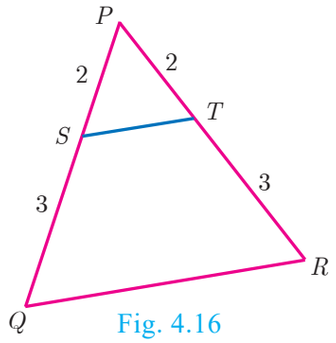


Fig. 4.16

Solution

(i) In ΔPST and ΔPQR,

$$\frac{PS}{PQ} = \frac{2}{2+1} = \frac{2}{3}, \frac{PT}{PR} = \frac{4}{4+2} = \frac{2}{3}$$

Thus, $\frac{PS}{PQ} = \frac{PT}{PR}$ and $\angle P$ is common

Therefore, by SAS similarity,

$\Delta PST \sim \Delta PQR$

(ii) In ΔPST and ΔPQR,

$$\frac{PS}{PQ} = \frac{2}{2+3} = \frac{2}{5}, \frac{PT}{PR} = \frac{2}{2+3} = \frac{2}{5}$$

Thus, $\frac{PS}{PQ} = \frac{PT}{PR}$ and $\angle P$ is common

Therefore, by SAS similarity,

$\Delta PST \sim \Delta PQR$

Example 4.2 Is $\Delta ABC \sim \Delta PQR$?

Solution In ΔABC and ΔPQR,

$$\frac{PQ}{AB} = \frac{3}{6} = \frac{1}{2}; \frac{QR}{BC} = \frac{4}{10} = \frac{2}{5}$$

since $\frac{1}{2} \neq \frac{2}{5}, \frac{PQ}{AB} \neq \frac{QR}{BC}$.

The corresponding sides are not proportional.

Therefore ΔABC is not similar to ΔPQR.

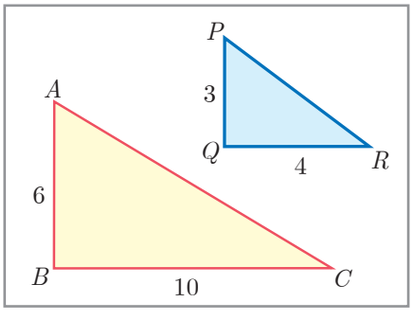


Fig. 4.17

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